Homework 6 Solutions

BSTA 551

Problem 1: Statistical vs. Practical Significance (Lesson 16)

A large randomized controlled trial ($n = 3{,}200$ patients per arm) evaluates a new lipid-lowering drug. The primary outcome is LDL cholesterol (mg/dL) measured after 6 months. Results:

- Control arm: $\bar{x}_{\text{control}} = 142.1$ mg/dL, $s_{\text{control}} = 28.4$ mg/dL
- Treatment arm: $\bar{x}_{\text{treat}} = 140.3$ mg/dL, $s_{\text{treat}} = 27.9$ mg/dL

For simplicity, assume equal and known $\sigma = 28.15$ mg/dL (pooled) and use a two-sample z-test.

(a) Compute the observed z-statistic and two-sided p-value for $H_0 : \mu_{\text{treat}} = \mu_{\text{control}}$ vs $H_a : \mu_{\text{treat}} \neq \mu_{\text{control}}$. Is the result statistically significant at $\alpha = 0.05$?

(b) Compute a 95% confidence interval for the difference in means $\mu_{\text{treat}} - \mu_{\text{control}}$.

(c) Compute Cohen's $d = |\bar{x}_{\text{treat}} - \bar{x}_{\text{control}}|/\sigma$. Using the table from class slides, how would you characterize the magnitude of this effect?

(d) Clinical guidelines consider an LDL reduction of ≥ 10 mg/dL to be clinically meaningful. Based on your confidence interval from part (b), does this trial provide evidence of a clinically meaningful reduction? Explain clearly, referencing both the interval and the clinical threshold.

(e) In 2–3 sentences, explain how a study can produce a statistically significant result that is not clinically significant, and why this distinction matters for medical decision-making.

Part (a) — z-statistic and p-value

```
xbar_c <- 142.1; xbar_t <- 140.3
sigma   <- 28.15
n       <- 3200

se_diff <- sigma * sqrt(2 / n)
z_obs   <- (xbar_t - xbar_c) / se_diff
p_val   <- 2 * pnorm(-abs(z_obs))

cat("SE of difference:", round(se_diff, 4), "\n")
```

```
SE of difference: 0.7038
```

```
cat("z-statistic:     ", round(z_obs, 4), "\n")
```

```
z-statistic:      -2.5577
```

```
cat("Two-sided p-value:", round(p_val, 5), "\n")
```

```
Two-sided p-value: 0.01054
```

```
cat("Significant at alpha = 0.05?", p_val < 0.05, "\n")
```

```
Significant at alpha = 0.05? TRUE
```

The difference is $\bar{x}_{\text{treat}} - \bar{x}_{\text{control}} = 140.3 - 142.1 = -1.8$ mg/dL. With $\text{SE} = \sigma\sqrt{2/n} = 0.7038$, the z-statistic is $z = -2.5577$ and the two-sided p-value is $p = 0.01054$, which is less than 0.05. The result is **statistically significant**.

Part (b) — 95% CI for difference in means

```
z_crit <- qnorm(0.975)
ci_lo  <- (xbar_t - xbar_c) - z_crit * se_diff
ci_hi  <- (xbar_t - xbar_c) + z_crit * se_diff
cat("95% CI: [", round(ci_lo, 3), ",", round(ci_hi, 3), "] mg/dL\n")
```

```
95% CI: [ -3.179 , -0.421 ] mg/dL
```

The 95% CI for $\mu_{\text{treat}} - \mu_{\text{control}}$ is $(-3.18, -0.42)$ mg/dL.

Part (c) — Cohen's d

```r
d_cohen <- abs(xbar_t - xbar_c) / sigma
cat("Cohen's d:", round(d_cohen, 4), "\n")
```

```
Cohen's d: 0.0639
```

Cohen's $d = 1.8/28.15 \approx 0.064$, which is well below 0.2. This is classified as a **negligible** effect by the standard Cohen's d table.

Part (d) — Clinical significance

The entire 95% CI $(-3.18, -0.42)$ mg/dL lies far above the clinical threshold of -10 mg/dL (a 10 mg/dL reduction). The upper end of the CI is positive, meaning the data are compatible with essentially no reduction or even a slight increase. There is **no evidence of a clinically meaningful LDL reduction**: the entire plausible range of effects falls well within the negligible range, far from the 10 mg/dL threshold.

Part (e) — Conceptual

With very large sample sizes, even negligible true effects produce extremely small standard errors, making the test statistic large enough to exceed any fixed critical value. Statistical significance measures whether an effect is *detectable* given the sample size — not whether it is *large enough to matter*. A drug that reduces LDL by 1.8 mg/dL in 3,200 patients per arm would likely not be approved or prescribed, since the cost, inconvenience, and potential side effects cannot be justified by a sub-clinical benefit. Reporting only the p-value obscures this; the CI makes the practical picture transparent.

Problem 2: CI-Test Duality — Theory and Verification (Lesson 16)

Let $X_1, \dots, X_n \overset{iid}{\sim} \text{Poisson}(\lambda)$. The exact two-sided $100(1-\alpha)\%$ confidence interval for λ based on $T = \sum_{i=1}^n X_i$ is:

$$\left(\frac{\chi^2_{\alpha/2,\, 2T}}{2n}, \frac{\chi^2_{1-\alpha/2,\, 2T+2}}{2n} \right)$$

(a) State the CI-Test Duality Theorem precisely (as given in Devore 9.6 / Lesson 16). Then use it to argue that the Poisson interval above corresponds to a level-α test of $H_0 : \lambda = \lambda_0$ vs $H_a : \lambda \neq \lambda_0$.

(b) A radiation monitoring station records gamma-ray events per 10-minute period. In $n = 8$ periods, the observed counts are $X = \{3, 7, 5, 4, 6, 8, 4, 5\}$.

(i) Compute the 95% Poisson CI using `qchisq()`.
(ii) Test $H_0 : \lambda = 4$ at $\alpha = 0.05$ using `poisson.test()`.
(iii) Test $H_0 : \lambda = 6$ at $\alpha = 0.05$.
(iv) Verify that each test decision is consistent with CI membership (duality check).

(c) State the 95% lower confidence bound dual to $H_0 : \lambda \leq \lambda_0$ vs $H_a : \lambda > \lambda_0$. Is there evidence that $\lambda > 4$?

Part (a) — Duality, theory

CI-Test Duality Theorem (Devore 9.6): Let $C(\mathbf{X})$ be a $100(1-\alpha)\%$ confidence interval for θ obtained by inverting a level-α test. Then:

$$\theta_0 \notin C(\mathbf{X}) \iff \text{reject } H_0 : \theta = \theta_0 \text{ at level } \alpha$$

$$\theta_0 \in C(\mathbf{X}) \iff \text{fail to reject } H_0 : \theta = \theta_0 \text{ at level } \alpha$$

The Poisson CI is constructed by inverting the exact Poisson test — it is the set of all λ_0 values for which the exact test fails to reject at level α. By the duality theorem, $\lambda_0 \in C(\mathbf{X})$ if and only if we fail to reject $H_0 : \lambda = \lambda_0$ at level α; the CI and the test convey identical information about which null values are consistent with the data.

Part (b) — Duality, computation

```r
x_counts  <- c(3, 7, 5, 4, 6, 8, 4, 5)
n_periods <- length(x_counts)
T_sum     <- sum(x_counts)
alpha_p   <- 0.05

cat("n =", n_periods, ", sum =", T_sum, ", xbar =", mean(x_counts), "\n\n")
```

```
n = 8 , sum = 42 , xbar = 5.25
```

```r
# (i) 95% Poisson CI via chi-squared formula
ci_lo_pois <- qchisq(alpha_p / 2,     df = 2 * T_sum)       / (2 * n_periods)
ci_hi_pois <- qchisq(1 - alpha_p / 2, df = 2 * T_sum + 2)   / (2 * n_periods)
cat("95% Poisson CI for lambda: [", round(ci_lo_pois, 4), ",",
    round(ci_hi_pois, 4), "]\n\n")
```

```
95% Poisson CI for lambda: [ 3.7837 , 7.0965 ]
```

```r
# (ii) Test H0: lambda = 4
test_4 <- poisson.test(T_sum, T = n_periods, r = 4, alternative = "two.sided")
cat("H0: lambda = 4\n")
```

```
H0: lambda = 4
```

```r
cat("  p-value:", round(test_4$p.value, 4), "\n")
```

```
  p-value: 0.0918
```

```r
cat("  Decision:", ifelse(test_4$p.value < 0.05, "Reject H0", "Fail to reject H0"), "\n")
```

```
  Decision: Fail to reject H0
```

```r
cat("  lambda0 = 4 in CI?", ci_lo_pois <= 4 & 4 <= ci_hi_pois, "\n\n")
```

```
  lambda0 = 4 in CI? TRUE
```

```r
# (iii) Test H0: lambda = 6
test_6 <- poisson.test(T_sum, T = n_periods, r = 6, alternative = "two.sided")
cat("H0: lambda = 6\n")
```

```
H0: lambda = 6
```

```r
cat("  p-value:", round(test_6$p.value, 4), "\n")
```

```
  p-value: 0.4271
```

```r
cat("  Decision:", ifelse(test_6$p.value < 0.05, "Reject H0", "Fail to reject H0"), "\n")
```

```
  Decision: Fail to reject H0
```

```r
cat("  lambda0 = 6 in CI?", ci_lo_pois <= 6 & 6 <= ci_hi_pois, "\n")
```

```
  lambda0 = 6 in CI? TRUE
```

(iv) Duality check:

- $\lambda_0 = 4$: test fails to reject H_0 ($p = 0.0918$); $\lambda_0 = 4$ is inside the CI. **Consistent**
- $\lambda_0 = 6$: test fails to reject H_0 ($p = 0.4271$); $\lambda_0 = 6$ is inside the CI. **Consistent**

Part (c) — One-sided duality

By the one-sided duality theorem, the dual lower bound for $H_0 : \lambda \leq \lambda_0$ vs $H_a : \lambda > \lambda_0$ is:

$$\underline{\lambda} = \frac{\chi^2_{\alpha,\, 2T}}{2n}$$

```r
lb_one  <- qchisq(alpha_p, df = 2 * T_sum) / (2 * n_periods)
test_gt <- poisson.test(T_sum, T = n_periods, r = 4, alternative = "greater")
cat("95% lower confidence bound:", round(lb_one, 4), "\n")
```

```
95% lower confidence bound: 3.9923
```

```r
cat("Lower bound > 4?", lb_one > 4, "\n")
```

```
Lower bound > 4? FALSE
```

```r
cat("One-sided p-value (Ha: lambda > 4):", round(test_gt$p.value, 4), "\n")
```

```
One-sided p-value (Ha: lambda > 4): 0.0512
```

The lower bound is $\underline{\lambda} = 3.9923$. Since $3.9923 \quad 4$, we fail to reject $H_0 : \lambda \leq 4$ at the 5% level.

Problem 3: Constructing CIs by Inverting Tests (Lesson 16)

A medical device manufacturer tests the precision of a glucose sensor. FDA requirements are $\sigma < 8$ mg/dL (i.e., $\sigma^2 < 64$). A quality control study with $n = 20$ measurements yields $s^2 = 42.3$ (mg/dL)^2.

(a) Starting from $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, derive the two-sided $100(1-\alpha)\%$ CI for σ^2 by inverting the acceptance region. Show each algebraic step.

(b) Compute the 95% CI for σ^2 and σ.

(c) To test $H_0 : \sigma^2 \geq 64$ vs $H_a : \sigma^2 < 64$ at $\alpha = 0.05$, compute the 95% upper confidence bound for σ^2 and state your conclusion.

(d) Verify by computing the chi-squared test p-value directly. Does the decision match?

```r
set.seed(551)
los_data <- rexp(30, rate = 0.18)
```

Part (a) — Derivation

The pivot is $Q = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. The acceptance region for testing $H_0 : \sigma^2 = \sigma_0^2$ is:

$$\chi^2_{1-\alpha/2,\, n-1} \leq \frac{(n-1)s^2}{\sigma_0^2} \leq \chi^2_{\alpha/2,\, n-1}$$

Inverting — solve for σ_0^2. Divide through (noting that dividing by a positive σ_0^2 preserves the direction, then inverting the chi-squared bounds flips the inequalities):

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\, n-1}} \leq \sigma_0^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\, n-1}}$$

This is the $100(1-\alpha)\%$ two-sided CI for σ^2. Note that $\chi^2_{\alpha/2} < \chi^2_{1-\alpha/2}$, so the larger denominator in the lower bound gives a smaller value, as required. $\square$

Part (b) — Numeric CI

```r
n_dev  <- 20; s2_dev <- 42.3; df_dev <- n_dev - 1; alpha_d <- 0.05

chi_upper <- qchisq(1 - alpha_d/2, df = df_dev)  # larger quantile -> smaller lower bound
chi_lower <- qchisq(    alpha_d/2, df = df_dev)  # smaller quantile -> larger upper bound

ci_v_lo <- df_dev * s2_dev / chi_upper
ci_v_hi <- df_dev * s2_dev / chi_lower

cat("chi^2 quantiles:", round(chi_lower, 3), "(lower),", round(chi_upper, 3), "(upper)\n")
```

```
chi^2 quantiles: 8.907 (lower), 32.852 (upper)
```

```r
cat("95% CI for sigma^2: [", round(ci_v_lo, 3), ",", round(ci_v_hi, 3), "]\n")
```

```
95% CI for sigma^2: [ 24.464 , 90.237 ]
```

```r
cat("95% CI for sigma:   [", round(sqrt(ci_v_lo), 3), ",", round(sqrt(ci_v_hi), 3), "]\n")
```

```
95% CI for sigma:   [ 4.946 , 9.499 ]
```

Part (c) — One-sided upper bound

For $H_0 : \sigma^2 \geq 64$ vs $H_a : \sigma^2 < 64$ (lower-tailed), the dual 95% upper confidence bound is:

$$\bar{\sigma}^2 = \frac{(n-1)s^2}{\chi^2_{\alpha,\, n-1}}$$

```r
chi_one  <- qchisq(alpha_d, df = df_dev)   # lower-tail alpha quantile
ub_var   <- df_dev * s2_dev / chi_one
sigma0_sq <- 64

cat("95% upper confidence bound for sigma^2:", round(ub_var, 3), "\n")
```

```
95% upper confidence bound for sigma^2: 79.44
```

```r
cat("Upper bound < 64?", ub_var < sigma0_sq, "\n")
```

```
Upper bound < 64? FALSE
```

```r
cat("Conclusion:", ifelse(ub_var < sigma0_sq,
    "Reject H0: sigma^2 >= 64. Sensor meets FDA standard.",
    "Fail to reject H0."), "\n")
```

```
Conclusion: Fail to reject H0.
```

The 95% upper bound is 79.44. Since 79.44   64, we fail to reject H_0 and conclude the sensor does not meet the precision requirement.

Part (d) — Direct chi-squared test

```r
chi_obs <- df_dev * s2_dev / sigma0_sq
p_chi   <- pchisq(chi_obs, df = df_dev, lower.tail = TRUE)  # lower-tailed

cat("chi^2_obs:", round(chi_obs, 4), "\n")
```

```
chi^2_obs: 12.5578
```

```r
cat("p-value (lower tail):", round(p_chi, 4), "\n")
```

```
p-value (lower tail): 0.1396
```

```r
cat("Reject H0?", p_chi < alpha_d, "-- consistent with part (c)?",
    (p_chi < alpha_d) == (ub_var < sigma0_sq), "\n")
```

```
Reject H0? FALSE -- consistent with part (c)? TRUE
```

The direct test confirms the CI-based conclusion: $p = 0.1396$ 0.05, so we fail to reject H_0 — **consistent with the upper bound in (c)**.

Problem 4: Multiple Testing and Simultaneous CIs (Lesson 16)

A Phase III clinical trial evaluates a new heart failure treatment on $m = 6$ pre-specified endpoints. You are given estimated treatment effects $\hat{\delta}_j$ and standard errors $\widehat{SE}_j$:

Endpoint	$\hat{\delta}_j$	$\widehat{SE}_j$
6-min walk distance (meters)	28.4	9.1
NT-proBNP (pg/mL, log-scale)	-0.31	0.09
Quality of life score (0–100)	5.2	2.1
Systolic BP (mmHg)	-4.8	2.3
Heart rate (bpm)	-3.1	1.6
Ejection fraction (%)	2.9	1.8

Assume $\hat{\delta}_j/\widehat{SE}_j \approx Z_j \sim N(0,1)$ under H_{0j}.

(a) For each endpoint, compute the unadjusted z-test p-value, 95% CI, and rejection decision at $\alpha = 0.05$.

(b) Apply Bonferroni correction to control the familywise error rate at 5%. Which endpoints remain significant?

(c) Construct Bonferroni-adjusted 95% simultaneous CIs. Using the duality theorem, explain why wider intervals achieve simultaneous coverage.

(d) Under the global null with 6 independent tests at $\alpha = 0.05$, compute the exact probability of at least one false rejection and the Bonferroni upper bound.

```r
endpoints <- c("6-min walk (m)", "NT-proBNP (log)", "QoL score",
               "Systolic BP (mmHg)", "Heart rate (bpm)", "EF (%)")
delta_hat <- c(28.4, -0.31, 5.2, -4.8, -3.1, 2.9)
se_hat    <- c(9.1,   0.09, 2.1,  2.3,  1.6,  1.8)
m         <- length(endpoints)
alpha_mf  <- 0.05
```

Part (a) — Unadjusted tests and CIs

```r
z_stats    <- delta_hat / se_hat
p_unadj    <- 2 * pnorm(-abs(z_stats))
z_95       <- qnorm(0.975)

ci_lo_ind <- delta_hat - z_95 * se_hat
ci_hi_ind <- delta_hat + z_95 * se_hat

tibble(
  Endpoint         = endpoints,
  `delta-hat`      = round(delta_hat, 2),
  SE               = round(se_hat, 2),
  `z-stat`         = round(z_stats, 3),
  `p-value`        = round(p_unadj, 4),
  `Reject (5%)`    = ifelse(p_unadj < alpha_mf, "Yes", "No"),
  `95% CI`         = sprintf("[%.2f, %.2f]", ci_lo_ind, ci_hi_ind)
) |> knitr::kable(align = "lrrrrlc")
```

Endpoint	delta-hat	SE	z-stat	p-value	Reject (5%)	95% CI
6-min walk (m)	28.40	9.10	3.121	0.0018	Yes	[10.56, 46.24]
NT-proBNP (log)	-0.31	0.09	-3.444	0.0006	Yes	[-0.49, -0.13]
QoL score	5.20	2.10	2.476	0.0133	Yes	[1.08, 9.32]
Systolic BP (mmHg)	-4.80	2.30	-2.087	0.0369	Yes	[-9.31, -0.29]
Heart rate (bpm)	-3.10	1.60	-1.938	0.0527	No	[-6.24, 0.04]
EF (%)	2.90	1.80	1.611	0.1072	No	[-0.63, 6.43]

Part (b) — Bonferroni correction

```r
alpha_bon <- alpha_mf / m
z_bon     <- qnorm(1 - alpha_bon / 2)

cat("Bonferroni alpha*:", round(alpha_bon, 5), "\n")
```

```
Bonferroni alpha*: 0.00833
```

```r
cat("Critical z value: ", round(z_bon, 4), "\n\n")
```

```
Critical z value:  2.6383
```

```r
cat("Endpoints significant after Bonferroni:\n")
```

```
Endpoints significant after Bonferroni:
```

```r
cat(paste(endpoints[p_unadj < alpha_bon], collapse = "\n"), "\n")
```

```
6-min walk (m)
NT-proBNP (log)
```

At the Bonferroni-corrected level $\alpha^* = 0.00833$ (z-critical = 2.6383), 2 endpoint(s) remain significant: **6-min walk (m), NT-proBNP (log)**.

Part (c) — Bonferroni simultaneous CIs

```r
ci_lo_bon <- delta_hat - z_bon * se_hat
ci_hi_bon <- delta_hat + z_bon * se_hat

tibble(
  Endpoint             = endpoints,
  `Indiv. 95% CI`      = sprintf("[%.2f, %.2f]", ci_lo_ind, ci_hi_ind),
  `Bonf. Simult. CI`   = sprintf("[%.2f, %.2f]", ci_lo_bon, ci_hi_bon)
) |> knitr::kable()
```

Endpoint	Indiv. 95% CI	Bonf. Simult. CI
6-min walk (m)	[10.56, 46.24]	[4.39, 52.41]
NT-proBNP (log)	[-0.49, -0.13]	[-0.55, -0.07]
QoL score	[1.08, 9.32]	[-0.34, 10.74]
Systolic BP (mmHg)	[-9.31, -0.29]	[-10.87, 1.27]
Heart rate (bpm)	[-6.24, 0.04]	[-7.32, 1.12]
EF (%)	[-0.63, 6.43]	[-1.85, 7.65]

The Bonferroni simultaneous CIs are wider than the individual 95% CIs. **By duality**, a $100(1-\alpha)\%$ CI is the set of θ_0 values that a level-α test fails to reject. To ensure that *all* m CIs simultaneously cover their true parameters with probability $\geq 1-\alpha$, each individual test must run at level $\alpha^* = \alpha/m$. A more stringent test level means a wider acceptance region, which inverts to a wider CI — simultaneous coverage is achieved by widening each interval proportionally.

Part (d) — Familywise error rate

```
fwer_exact <- 1 - (1 - alpha_mf)^m
fwer_bon   <- m * alpha_mf

cat("FWER exact (independence):", round(fwer_exact, 4), "\n")
```

```
FWER exact (independence): 0.2649
```

```
cat("Bonferroni upper bound:   ", round(fwer_bon, 4), "\n")
```

```
Bonferroni upper bound:    0.3
```

Under the global null and independence:

$$P(\text{at least one false rejection}) = 1 - (1 - 0.05)^6 = 0.2649$$

The Bonferroni bound is $m\alpha = 6 \times 0.05 = 0.30$, which is conservative (a union bound) and slightly exceeds the exact value.

Problem 5: Bootstrap Hypothesis Testing (Lesson 17)

Serum ferritin (ng/mL) is a biomarker for iron stores. A study of $n = 18$ CKD patients gives:

```
ferritin <- c(28.4, 45.2, 19.8, 67.3, 31.5, 88.1, 22.6, 54.9,
              41.2, 76.8, 33.7, 29.4, 58.3, 44.1, 92.5, 37.6,
              25.8, 61.4)
```

The population norm is $\mu_0 = 40$ ng/mL. Test $H_0 : \mu = 40$ vs $H_a : \mu \neq 40$ at $\alpha = 0.05$.

(a) Make a histogram and describe the distribution's shape. Is normality reasonable?

(b) Compute $T_{\text{obs}} = (\bar{x} - \mu_0)/(s/\sqrt{n})$ and the parametric t-test p-value. State your conclusion.

(c) Implement the **correct** bootstrap test with $B = 5{,}000$ (`set.seed(551)`): (i) shift data to H_0, (ii) compute bootstrap t-statistics, (iii) compute p-value as $\hat{p} = B^{-1} \sum \mathbf{1}(|T^{*b}| \geq |T_{\text{obs}}|)$, (iv) compare to the t-test.

(d) Implement the **naive (incorrect)** bootstrap (resample from original, compare $|\bar{x}^{*b} - \bar{x}|$ to $|\bar{x} - \mu_0|$). Compute the naive p-value and explain in 2–3 sentences why this approach is invalid.

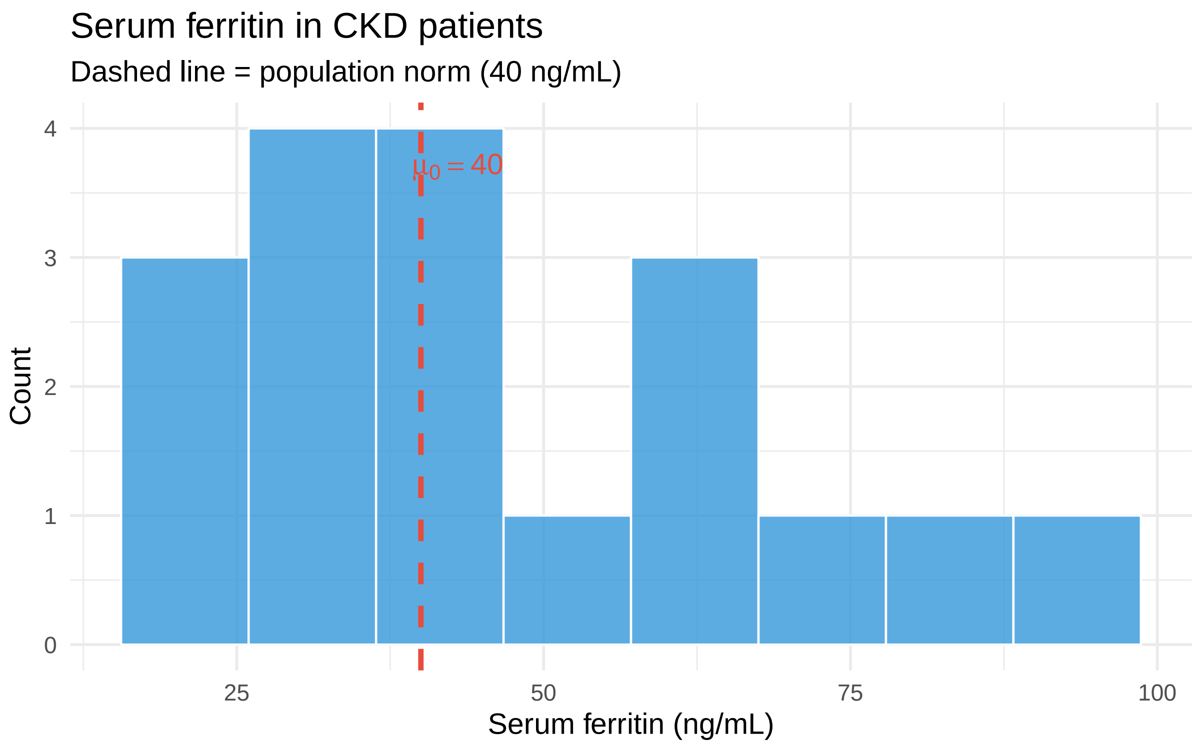
(e) Plot histograms of both bootstrap distributions side by side, marking the observed statistic and p-value region on each.

(f) Would you expect the bootstrap percentile CI (from resampling the original data) to be approximately dual to the correct bootstrap test from (c)? Explain, referencing the pivot-matching discussion in Lesson 17.

```r
mu_0 <- 40
n_f  <- length(ferritin)
xbar_f <- mean(ferritin); s_f <- sd(ferritin); se_f <- s_f / sqrt(n_f)
```

Part (a) — Histogram and shape

```r
ggplot(tibble(x = ferritin), aes(x = x)) +
  geom_histogram(bins = 8, fill = "#3498db", color = "white", alpha = 0.8) +
  geom_vline(xintercept = mu_0, linetype = "dashed", color = "#e74c3c", linewidth = 1.2) +
  annotate("text", x = mu_0 + 3, y = 3.7, label = expression(mu[0] == 40),
           color = "#e74c3c", size = 5) +
  labs(title = "Serum ferritin in CKD patients",
       subtitle = "Dashed line = population norm (40 ng/mL)",
       x = "Serum ferritin (ng/mL)", y = "Count")
```



The distribution is **right-skewed** with a long upper tail (values up to 92.5 ng/mL while many observations cluster below 45 ng/mL). With $n = 18$ and noticeable skewness, the normality assumption is questionable — a bootstrap test is a sensible choice here.

Part (b) — Parametric t-test

```
T_obs_f  <- (xbar_f - mu_0) / se_f
p_ttest  <- t.test(ferritin, mu = mu_0)$p.value

cat("Sample mean:", round(xbar_f, 3), "ng/mL\n")
```

```
Sample mean: 47.7 ng/mL
```

```
cat("Sample SD:  ", round(s_f, 3),   "ng/mL\n")
```

```
Sample SD:   22.281 ng/mL
```

```r
cat("T_obs:       ", round(T_obs_f, 4), "\n")
```

```
T_obs:       1.4662
```

```r
cat("t-test p:    ", round(p_ttest, 4), "\n")
```

```
t-test p:    0.1608
```

```r
cat("Decision:    ", ifelse(p_ttest < 0.05, "Reject H0", "Fail to reject H0"), "\n")
```

```
Decision:    Fail to reject H0
```

Part (c) — Correct bootstrap test

```r
set.seed(551); B_f <- 5000

ferritin_sh <- ferritin - xbar_f + mu_0   # (i) shift to H0

T_boot_f <- replicate(B_f, {              # (ii) bootstrap t-statistics
  bs <- sample(ferritin_sh, n_f, replace = TRUE)
  (mean(bs) - mu_0) / (sd(bs) / sqrt(n_f))
})

p_boot_f <- mean(abs(T_boot_f) >= abs(T_obs_f))   # (iii) p-value

# (iv) comparison
cat("Bootstrap p-value:", round(p_boot_f, 4), "\n")
```

```
Bootstrap p-value: 0.175
```

```r
cat("t-test p-value:    ", round(p_ttest,  4), "\n")
```

```
t-test p-value:    0.1608
```

```r
cat("Bootstrap decision:", ifelse(p_boot_f < 0.05, "Reject H0", "Fail to reject H0"), "\n")
```

```
Bootstrap decision: Fail to reject H0
```

Both methods fail to reject H_0, and the p-values are close — consistent with Lesson 17's simulation showing that differences between bootstrap and t-test shrink as n increases.

Part (d) — Naive (incorrect) bootstrap

```
set.seed(551)
T_naive_f <- replicate(B_f, mean(sample(ferritin, n_f, replace = TRUE)))
p_naive_f <- mean(abs(T_naive_f - xbar_f) >= abs(xbar_f - mu_0))
cat("Naive p-value:", round(p_naive_f, 4), "\n")
```

```
Naive p-value: 0.1396
```

The naive p-value is 0.1396. This approach is invalid because resampling from the original data produces a distribution centered at $\bar{x} = 47.7$ ng/mL — not at the null value $\mu_0 = 40$ ng/mL. A valid null distribution must represent how extreme the observed statistic would be *if H_0 were true*. Comparing variability around $\bar{x}$ answers a different question (how variable is our estimator?) rather than the question of interest (how unusual is our result under H_0?).

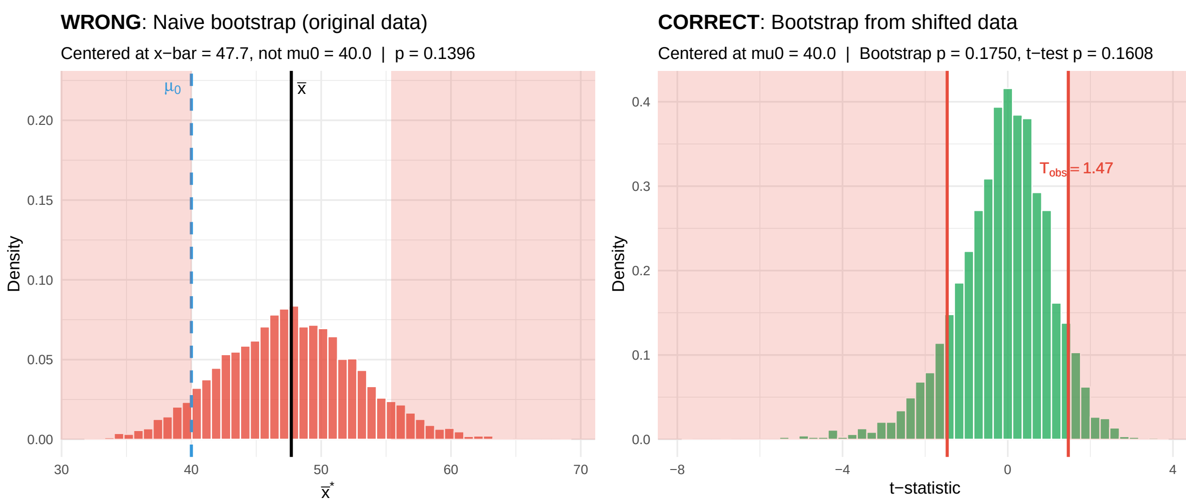
Part (e) — Side-by-side plots

```
p_nv <- ggplot(tibble(T = T_naive_f), aes(x = T)) +
  geom_histogram(aes(y = after_stat(density)), bins = 50,
                 fill = "#e74c3c", color = "white", alpha = 0.8) +
  geom_vline(xintercept = xbar_f, color = "black", linewidth = 1.2) +
  geom_vline(xintercept = mu_0,   color = "#3498db", linewidth = 1.2, linetype = "dashed") +
  annotate("rect",
           xmin = xbar_f + abs(xbar_f - mu_0), xmax = Inf, ymin = 0, ymax = Inf,
           fill = "#e74c3c", alpha = 0.2) +
  annotate("rect",
           xmin = -Inf, xmax = xbar_f - abs(xbar_f - mu_0), ymin = 0, ymax = Inf,
           fill = "#e74c3c", alpha = 0.2) +
  annotate("text", x = xbar_f + 0.8, y = 0.22,
           label = "bar(x)", parse = TRUE, size = 5) +
  annotate("text", x = mu_0 - 0.8, y = 0.22,
           label = "mu[0]", parse = TRUE, size = 5, color = "#3498db", hjust = 1) +
  labs(title = expression(bold("WRONG") * ": Naive bootstrap (original data)"),
       subtitle = sprintf("Centered at x-bar = %.1f, not mu0 = %.1f  |  p = %.4f",
                          xbar_f, mu_0, p_naive_f),
       x = expression(bar(x)^"*"), y = "Density")

p_cr <- ggplot(tibble(T = T_boot_f), aes(x = T)) +
  geom_histogram(aes(y = after_stat(density)), bins = 50,
```

```r
                  fill = "#27ae60", color = "white", alpha = 0.8) +
  geom_vline(xintercept = c(-abs(T_obs_f), abs(T_obs_f)),
               color = "#e74c3c", linewidth = 1.2) +
  annotate("rect", xmin =  abs(T_obs_f), xmax =  Inf, ymin = 0, ymax = Inf,
            fill = "#e74c3c", alpha = 0.2) +
  annotate("rect", xmin = -Inf, xmax = -abs(T_obs_f), ymin = 0, ymax = Inf,
            fill = "#e74c3c", alpha = 0.2) +
  annotate("text", x = abs(T_obs_f) + 0.2, y = 0.32,
            label = bquote(T[obs] == .(round(T_obs_f, 2))),
            color = "#e74c3c", size = 4.5) +
  labs(title = expression(bold("CORRECT") * ": Bootstrap from shifted data"),
       subtitle = sprintf("Centered at mu0 = %.1f  |  Bootstrap p = %.4f, t-test p = %.4f",
                          mu_0, p_boot_f, p_ttest),
       x = "t-statistic", y = "Density")

p_nv + p_cr
```



Part (f) — Conceptual: CI-test duality

A bootstrap **percentile CI** (resampling from original data, using $\bar{X}^*$ as pivot) and the **bootstrap t-test** in (c) (using standardised $T^* = (\bar{X}^* - \mu_0)/(S^*/\sqrt{n})$ as pivot) use **different pivots**. As demonstrated in Lesson 17, duality is approximate only when both the CI and the test invert the *same* pivot. With the percentile CI (raw-mean pivot) vs. the standardised t-test, there will be systematic disagreements near the CI boundary — particularly evident with small, skewed samples like this one. To achieve approximate duality, one would need to use the raw mean difference $\bar{x}^* - \mu_0$ as the test statistic (matching the percentile CI's pivot), or use a bootstrap t-CI (percentile CI of the t-statistic) paired with the t-based bootstrap test.

Problem 6: Devore 9.6 (Modified)

The plasma concentration (mg/L) of a drug is measured for $n = 12$ patients:

$$X = \{1.8,\ 2.6,\ 3.1,\ 2.2,\ 4.7,\ 2.9,\ 3.8,\ 2.4,\ 1.6,\ 3.5,\ 4.1,\ 2.7\}$$

The regulatory target mean concentration is $\mu_0 = 3.0$ mg/L.

(a) Compute $\bar{x}$, s, and the 95% two-sided t-interval for μ.

(b) Use the duality theorem to determine whether you would reject $H_0 : \mu = 3.0$ at $\alpha = 0.05$ without computing a formal test statistic.

(c) Test $H_0 : \mu = 2.5$ vs $H_a : \mu > 2.5$ at $\alpha = 0.05$. Compute the t-statistic, p-value, and the dual one-sided 95% lower confidence bound. Do they agree?

(d) Sweep μ_0 over $[1.5, 4.5]$ in steps of 0.05. For each value compute the two-sided t-test p-value. Plot the p-value function and verify that the 95% CI corresponds exactly to the region where $p(\mu_0) \geq 0.05$.

```r
drug_conc <- c(1.8, 2.6, 3.1, 2.2, 4.7, 2.9, 3.8, 2.4, 1.6, 3.5, 4.1, 2.7)
n_dc      <- length(drug_conc)
xbar_dc   <- mean(drug_conc)
s_dc      <- sd(drug_conc)
se_dc     <- s_dc / sqrt(n_dc)
df_dc     <- n_dc - 1
```

Part (a) — Summary statistics and 95% t-interval

```r
t_crit_dc <- qt(0.975, df = df_dc)
ci_lo_dc  <- xbar_dc - t_crit_dc * se_dc
ci_hi_dc  <- xbar_dc + t_crit_dc * se_dc

cat("n:", n_dc, "  xbar:", round(xbar_dc, 4), "  s:", round(s_dc, 4), "\n")
```

```
n: 12   xbar: 2.95   s: 0.9357
```

```r
cat("95% t-CI: [", round(ci_lo_dc, 4), ",", round(ci_hi_dc, 4), "] mg/L\n")
```

```
95% t-CI: [ 2.3555 , 3.5445 ] mg/L
```

$\bar{x} = 2.95$, $s = 0.936$, 95% CI: $[2.356, 3.544]$ mg/L.

Part (b) — Duality conclusion for $\mu_0 = 3.0$

$\mu_0 = 3.0$ falls **inside** the 95% CI $[2.356, 3.544]$. By the CI-Test Duality Theorem, we therefore **fail to reject** $H_0 : \mu = 3.0$ at $\alpha = 0.05$ — no test statistic needed.

Part (c) — One-sided test and dual bound

```r
mu0_one   <- 2.5
T_obs_one <- (xbar_dc - mu0_one) / se_dc
p_one     <- pt(T_obs_one, df = df_dc, lower.tail = FALSE)  # Ha: mu > 2.5

t_one     <- qt(0.95, df = df_dc)   # one-sided critical value
lb_dc     <- xbar_dc - t_one * se_dc

cat("T_obs:", round(T_obs_one, 4), "\n")
```

```
T_obs: 1.666
```

```r
cat("One-sided p-value:", round(p_one, 4), "\n")
```

```
One-sided p-value: 0.0619
```

```r
cat("95% lower confidence bound:", round(lb_dc, 4), "\n")
```

```
95% lower confidence bound: 2.4649
```

```r
cat("Lower bound > 2.5?", lb_dc > mu0_one, "\n")
```

```
Lower bound > 2.5? FALSE
```

```r
cat("Duality consistent?", (p_one < 0.05) == (lb_dc > mu0_one), "\n")
```

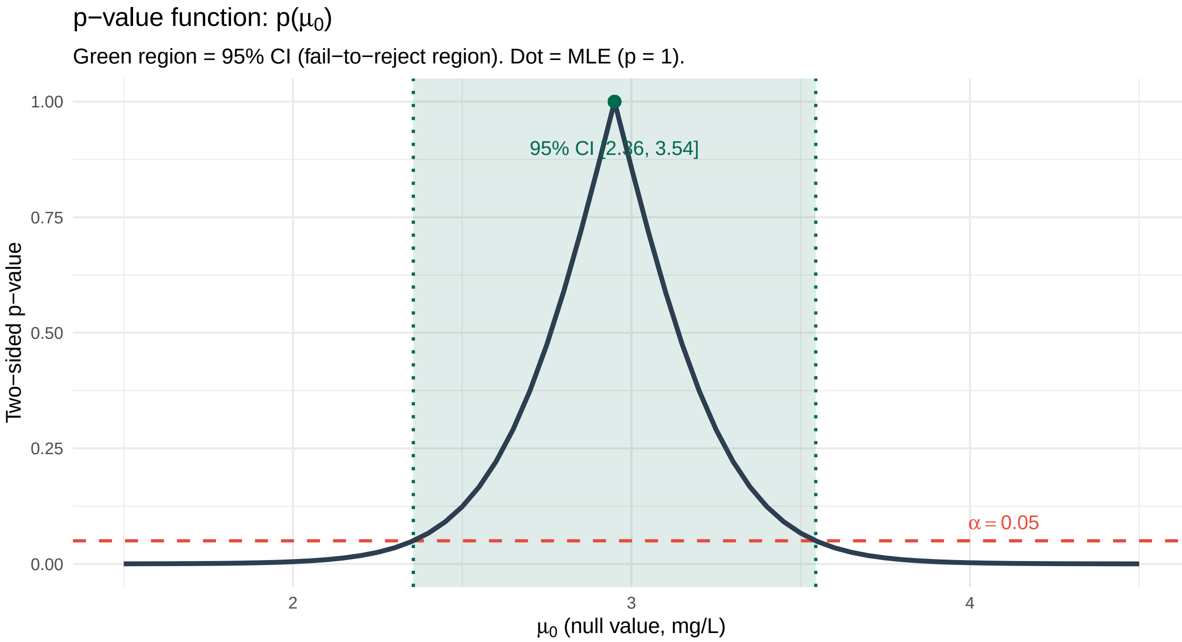
```
Duality consistent? TRUE
```

$T_{\text{obs}} = 1.666$, $p = 0.0619$. We **fail to reject** H_0 at $\alpha = 0.05$. The dual 95% lower bound is $\underline{\mu} = 2.4649$ mg/L; since 2.4649   2.5, the bound is consistent with failure to reject. **Duality confirmed**

Part (d) — p-value function

```r
mu0_grid <- seq(1.5, 4.5, by = 0.05)
pval_fn  <- map_dbl(mu0_grid, function(mu0) {
  t_stat <- (xbar_dc - mu0) / se_dc
  2 * pt(-abs(t_stat), df = df_dc)
})

ggplot(tibble(mu0 = mu0_grid, pval = pval_fn), aes(x = mu0, y = pval)) +
  annotate("rect", xmin = ci_lo_dc, xmax = ci_hi_dc, ymin = -Inf, ymax = Inf,
           fill = "#006a4e", alpha = 0.12) +
  geom_vline(xintercept = c(ci_lo_dc, ci_hi_dc),
             color = "#006a4e", linewidth = 1, linetype = "dotted") +
  geom_hline(yintercept = 0.05, linetype = "dashed", color = "#e74c3c", linewidth = 1) +
  geom_line(color = "#2c3e50", linewidth = 1.5) +
  geom_point(aes(x = xbar_dc, y = 1), size = 3.5, color = "#006a4e") +
  annotate("text", x = xbar_dc, y = 0.90,
           label = sprintf("95%% CI [%.2f, %.2f]", ci_lo_dc, ci_hi_dc),
           color = "#006a4e", size = 4.5) +
  annotate("text", x = 4.1, y = 0.09,
           label = expression(alpha == 0.05), color = "#e74c3c", size = 4.5) +
  labs(
    title = expression("p-value function: p(" * mu[0] * ")"),
    subtitle = "Green region = 95% CI (fail-to-reject region). Dot = MLE (p = 1).",
    x = expression(mu[0] * " (null value, mg/L)"),
    y = "Two-sided p-value"
  )
```



The green shaded region (where $p(\mu_0) \geq 0.05$) spans exactly $[2.356, 3.544]$ mg/L — matching the 95% t-CI from part (a) precisely. The CI is exactly the set of null values we would fail to reject at $\alpha = 0.05$, confirming the duality theorem visually.